

Implementation of STLC on an E-Commerce Website

Project Overview :

Suppose a company wants to develop an **E-Commerce website** where customers can:

- Register an account
- Log in
- Search for products
- Add products to the cart
- Place orders
- Make online payments
- Track orders

Phase 1: Requirement Analysis

Objective

- Understand what needs to be tested.

Activities

- The QA team receives the Software Requirement Specification (SRS) document.
- The tester studies the requirements and identifies all the features that need testing.

Features to Analyze

- User Registration
- Login
- Forgot Password
- Search Products
- Product Details
- Add to Cart
- Remove from Cart
- Wishlist
- Checkout

- Payment
- Order Confirmation
- Order Tracking
- Logout

Entry Criteria

- SRS document available.
- Requirements approved.

Exit Criteria

- All requirements understood.
- Testable requirements identified.
- Doubts clarified.

Deliverables

- Requirement Analysis Document
 - Requirement Traceability Matrix (RTM)
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Phase 2: Test Planning

Objective

- Plan how testing will be performed.

Activities

- The Test Lead prepares the Test Plan.

The testing team decides:

- Testing Scope
- Testing Types
- Team Members
- Schedule
- Risks
- Tools

Example

Testing Types

- Functional Testing
- Smoke Testing
- Regression Testing
- Usability Testing

Resources

- 1 Test Lead
- 3 QA Engineers

Timeline

- 20 Days

Entry Criteria

- Requirements finalized.

Exit Criteria

- Test Plan approved.

Deliverables

- Test Plan
- Test Strategy
- Resource Plan

Phase 3: Test Case Development

Objective

- Prepare test cases for every feature.

Activities

- The QA team writes test cases for all modules.

Example Test Cases

Registration

- Verify successful registration.
- Verify duplicate email validation.
- Verify mandatory fields.

Login

- Verify valid login.
- Verify invalid password.
- Verify empty username.

Search

- Verify product search.
- Verify search with invalid keyword.

Cart

- Verify Add to Cart.
- Verify Remove from Cart.
- Verify quantity update.

Payment

- Verify successful payment.
- Verify failed payment.
- Verify payment cancellation.

Entry Criteria

- Test Plan completed.

Exit Criteria

- Test Cases completed.
- Test Data prepared.

Deliverables

- Test Cases
 - Test Data
 - RTM
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Phase 4: Test Environment Setup

Objective

- Prepare the testing environment.

Activities

- Install the application and configure all required systems.

Environment

- Operating System
- Windows 11
- Browser
- Google Chrome
- Database
- MySQL
- Server
- QA Server
- Test Accounts
- Customer Account
- Admin Account
- Payment Sandbox

Entry Criteria

- Build received.
- Test Cases ready.

Exit Criteria

- Environment configured.
- Application installed.
- Test accounts created.

Deliverables

- Environment Setup Report
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Phase 5: Test Execution

Objective

- Execute all prepared test cases.

Activities

- The tester executes every test case and compares the actual result with the expected result.

Example

Registration

- Expected
- User account should be created.
- Actual
- Pass

Login

- Expected
- User should login successfully.
- Actual
- Pass

Search

- Expected
- Relevant products should be displayed.
- Actual
- Pass

Add to Cart

- Expected
- Selected product should be added.
- Actual
- Pass

Payment

- Expected
- Order should be placed successfully.
- Actual
- Payment failed.
- Status
- Fail
- Bug Reported.

Entry Criteria

- Test Environment ready.
- Test Cases approved.

Exit Criteria

- Test Cases executed.
- Bugs reported.

Deliverables

- Test Execution Report
 - Defect Report
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Phase 6: Test Closure

Objective

- Complete testing activities and prepare reports.

Activities

- The QA team reviews testing results.

Test Summary

- Total Test Cases
200
- Passed
190

- Failed
10
- Blocked
0
- Critical Bugs
0
- Minor Bugs
2
- Regression Testing
Completed
- Testing Status
Completed

Documents Prepared

- Test Summary Report
- Defect Summary Report
- Lessons Learned
- Test Closure Report

Entry Criteria

- Testing completed.

Exit Criteria

- Reports prepared.
- Client approval received.

Deliverables

- Test Closure Report
 - Final Sign-Off
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